18179: Software Development

GE2B: Guided Exercise 2B

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Grammar Applied

<json> ::= {<data>}

<data> ::= <field1><seperator> <field2>
<seperator> ::= “,”

<field1> ::= <label_PROJECT_ID><equal><value_PROJECT_ID>

<label_PROJECT_ID> ::= <STRING_TOKEN>
<STRING_TOKEN> ::= “PROJECT_ID”

<equal> ::= “:”

<value_PROJECT_ID> ::= “<HEX32>”
<HEX32> ::= <HEX><HEX><HEX><HEX><HEX><HEX><HEX><HEX><HEX><HEX><HEX><HEX><HEX><HEX><HEX><HEX><HEX><HEX><HEX><HEX>
<HEX><HEX><HEX><HEX><HEX><HEX><HEX><HEX><HEX><HEX><HEX><HEX><HEX>
<HEX> ::= “0”| “1” | “2”| “3” | “4”| “5” | “6”| “7” | “8”| “9” | “a”| “b” | “c”| “d” | “e”| “f” | “A”| “B” | “C”| “D” | “E”| “F”

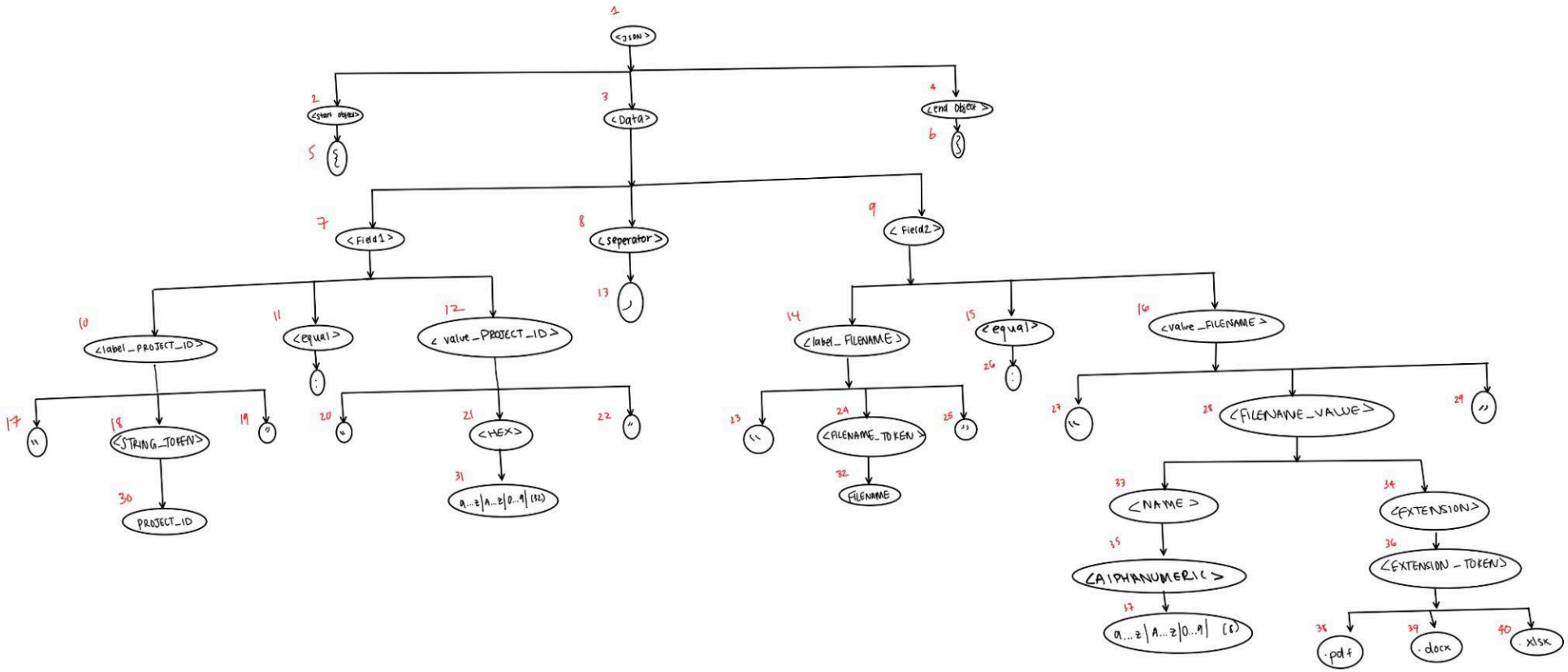
<field2> ::= <label_FILENAME><equal><value_FILENAME>
<equal> ::= “:”

<label_FILENAME> ::= <FILENAME_TOKEN>
<FILENAME_TOKEN> ::= “FILENAME”

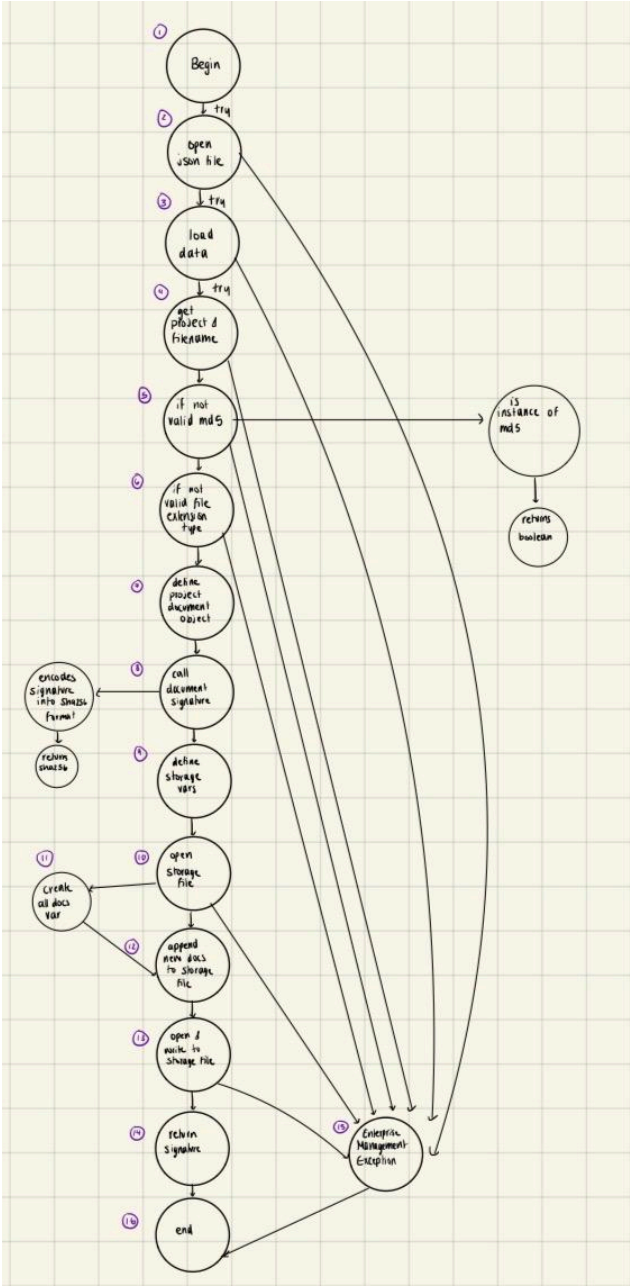
<value_FILENAME> ::= <FILENAME_VALUE>
<FILENAME_VALUE> ::= <NAME><EXTENSION>
<NAME> ::= <ALPHANUMERIC><ALPHANUMERIC><ALPHANUMERIC><ALPHANUMERIC><ALPHANUMERIC><ALPHANUMERIC>
<ALPHANUMERIC><ALPHANUMERIC>
<ALPHANUMERIC> ::= “A”| … | “Z” | “a” | … | “z” | “0” | … | “9”

<EXTENSION> ::= <EXTENSION_TOKEN>
<EXTENSION_TOKEN> ::= “.pdf” | “.docx” | “.xlsx”

1. Syntactic Analysis: Derivation Tree



2. Structural Testing: Control Flow Diagram



C = 23 - 16 + 2 = 9

Syntactic Analysis Test Cases

#	NODE	TERMINAL	TYPE (DUPLICATION / DELETION/ MODIFICATION / VALID)	ID TEST	DESCRIPTION	FILE PATH	FILE CONTENT	EXPECTED RESULT	OBSERVATIONS
1	1	NT	VALID	tc_sa_01_valid_baseline	Valid JSON – baseline (.pdf)	tc_sa_01.json	{ "PROJECT_ID": "0123456789abcdef0123456789abcdef", "FILENAME": "ABcd1234.pdf" }	Returns SHA-256 hash: c29b7d03db0e597a9a178318e0121697c8a2523c874190c1d5cfb352e0dcbd55 (use @freeze_time in test)	Baseline
2	1	NT	DELETION	tc_sa_02_json_empty	Entire JSON content deleted (empty file)	tc_sa_02.json		EnterpriseManagementException: JSON data has no valid values	Node 1 deleted
3	1	NT	DUPLICATION	tc_sa_03_json_duplicated	JSON object duplicated at root	tc_sa_03.json	{ "PROJECT_ID": "0123456789abcdef0123456789abcdef", "FILENAME": "ABcd1234.pdf" } { "PROJECT_ID": "0123456789abcdef0123456789abcdef", "FILENAME": "ABcd1234.pdf" }	EnterpriseManagementException: The file is not JSON formatted	Node 1 duplicated
4	1	NT	MODIFICATION	tc_sa_04_json_not_json	File content is plain text, not JSON	tc_sa_04.json	this is not json	EnterpriseManagementException: The file is not JSON formatted	Node 1 modified
5	2	NT	DELETION	tc_sa_05_open_brace_deleted	<open_object> deleted: opening { missing	tc_sa_05.json	"PROJECT_ID": "0123456789abcdef0123456789abcdef", "FILENAME": "ABcd1234.pdf" }	EnterpriseManagementException: The file is not JSON formatted	Node 2/5 deleted
6	2	NT	DUPLICATION	tc_sa_06_open_brace_duplicated	<open_object> duplicated: {{ at start	tc_sa_06.json	{{ "PROJECT_ID": "0123456789abcdef	EnterpriseManagementException: The file is not JSON	Node 2/5 duplicated

							0123456789abcdef" , "FILENAME": "ABcd1234.pdf" }	formatted	
7	2	NT	MODIFICATION	tc_sa_07_open_brace_modified	<open_object> modified: [instead of {	tc_sa_07.json	["PROJECT_ID": "0123456789abcdef 0123456789abcdef" , "FILENAME": "ABcd1234.pdf" }	EnterpriseManagementException: The file is not JSON formatted	Node 2/5 modified
8	4	NT	DELETION	tc_sa_08_close_brace_deleted	<end_object> deleted: closing } missing	tc_sa_08.json	{ "PROJECT_ID": "0123456789abcdef 0123456789abcdef" , "FILENAME": "ABcd1234.pdf"	EnterpriseManagementException: The file is not JSON formatted	Node 4/6 deleted
9	4	NT	DUPLICATION	tc_sa_09_close_brace_duplicated	<end_object> duplicated: }} at end	tc_sa_09.json	{ "PROJECT_ID": "0123456789abcdef 0123456789abcdef" , "FILENAME": "ABcd1234.pdf" }}	EnterpriseManagementException: The file is not JSON formatted	Node 4/6 duplicated
10	4	NT	MODIFICATION	tc_sa_10_close_brace_modified	<end_object> modified:] instead of }	tc_sa_10.json	{ "PROJECT_ID": "0123456789abcdef 0123456789abcdef" , "FILENAME": "ABcd1234.pdf"]	EnterpriseManagementException: The file is not JSON formatted	Node 4/6 modified
11	3	NT	DELETION	tc_sa_11_data_deleted	<Data> deleted: only braces remain	tc_sa_11.json	{}	EnterpriseManagementException: JSON does not have expected structure	Node 3 deleted
12	3	NT	DUPLICATION	tc_sa_12_data_duplicated	<Data> duplicated: all fields repeated twice	tc_sa_12.json	{ "PROJECT_ID": "0123456789abcdef 0123456789abcdef" , "FILENAME": "ABcd1234.pdf", "PROJECT_ID": "0123456789abcdef 0123456789abcdef" ,	EnterpriseManagementException: JSON data has no valid values	Node 3 duplicated

							"FILENAME": "ABcd1234.pdf" }		
13	7	NT	DELETION	tc_sa_13_field1_deleted	<Field1> deleted: no PROJECT_ID field	tc_sa_13.json	{ "FILENAME": "ABcd1234.pdf" }	EnterpriseManagementException: JSON does not have expected structure	Node 7 deleted
14	7	NT	DUPLICATION	tc_sa_14_field1_duplicated	<Field1> duplicated: PROJECT_ID field twice	tc_sa_14.json	{ "PROJECT_ID": "0123456789abcdef0123456789abcdef" , "PROJECT_ID": "0123456789abcdef0123456789abcdef" , "FILENAME": "ABcd1234.pdf" }	EnterpriseManagementException: JSON data has no valid values	Node 7 duplicated
15	8	NT	DELETION	tc_sa_15_separator_deleted	<separator> deleted: comma between fields missing	tc_sa_15.json	{ "PROJECT_ID": "0123456789abcdef0123456789abcdef" "FILENAME": "ABcd1234.pdf" }	EnterpriseManagementException: The file is not JSON formatted	Node 8/13 deleted
16	8	NT	DUPLICATION	tc_sa_16_separator_duplicated	<separator> duplicated: ,, between fields	tc_sa_16.json	{ "PROJECT_ID": "0123456789abcdef0123456789abcdef" ,, "FILENAME": "ABcd1234.pdf" }	EnterpriseManagementException: The file is not JSON formatted	Node 8/13 duplicated
17	8	NT	MODIFICATION	tc_sa_17_separator_modified	<separator> modified: ; instead of ,	tc_sa_17.json	{ "PROJECT_ID": "0123456789abcdef0123456789abcdef" ; "FILENAME": "ABcd1234.pdf" }	EnterpriseManagementException: The file is not JSON formatted	Node 8/13 modified
18	9	NT	DELETION	tc_sa_18_field2_deleted	<Field2> deleted: no FILENAME field	tc_sa_18.json	{ "PROJECT_ID": "0123456789abcdef0123456789abcdef" }	EnterpriseManagementException: JSON does not have expected structure	Node 9 deleted
19	9	NT	DUPLICATION	tc_sa_19_field2_duplicated	<Field2> duplicated: FILENAME field twice	tc_sa_19.json	{ "PROJECT_ID": "0123456789abcdef0123456789abcdef"	EnterpriseManagementException: JSON data has no valid values	Node 9 duplicated

							, "FILENAME": "ABcd1234.pdf", "FILENAME": "ABcd1234.pdf" }		

20	10	NT	DELETION	tc_sa_20_label_pid_deleted	<label_PROJECT_ID> deleted: key missing	tc_sa_20.json	{
							:
							"0123456789abcdef0123456789abcdef"
							,
							"FILENAME":
							"ABcd1234.pdf"
							}
								EnterpriseManagementException: The file is not JSON formatted	Node 10 deleted

21	10	NT	DUPLICATION	tc_sa_21_label_pid_duplicated	<label_PROJECT_ID> duplicated: key repeated	tc_sa_21.json	{
							"PROJECT_ID""PROJECT_ID":
							"0123456789abcdef0123456789abcdef"
							,
							"FILENAME":
							"ABcd1234.pdf"
							}
								EnterpriseManagementException: The file is not JSON formatted	Node 10 duplicated

22	10	NT	MODIFICATION	tc_sa_22_label_pid_modified	<label_PROJECT_ID> modified: wrong key name	tc_sa_22.json	{
							"PROJ_ID":
							"0123456789abcdef0123456789abcdef"
							,
							"FILENAME":
							"ABcd1234.pdf"
							}
								EnterpriseManagementException: JSON does not have expected structure	Node 10 modified

23	11	NT	DELETION	tc_sa_23_colon_pid_deleted	<equal>(Field1) deleted: colon after PROJECT_ID missing	tc_sa_23.json	{
							"PROJECT_ID"
							"0123456789abcdef0123456789abcdef"
							,
							"FILENAME":
							"ABcd1234.pdf"
							}
								EnterpriseManagementException: The file is not JSON formatted	Node 11/20 deleted

24	11	NT	DUPLICATION	tc_sa_24_colon_pid_duplicated	<equal>(Field1) duplicated: :: after PROJECT_ID	tc_sa_24.json	{
							"PROJECT_ID"::
							"0123456789abcdef0123456789abcdef"
							,
							"FILENAME":
							"ABcd1234.pdf"
							}
								EnterpriseManagementException: The file is not JSON formatted	Node 11/20 duplicated

25	11	NT	MODIFICATION	tc_sa_25_colon_pid_modified	<equal>(Field1) modified: = instead of :	tc_sa_25.json	{ "PROJECT_ID"= "0123456789abcdef0123456789abcdef"	EnterpriseManagementException: The file is not JSON formatted	Node 11/20 modified

							, "FILENAME": "ABcd1234.pdf" }		
26	12	NT	DELETION	tc_sa_26_value_pid_deleted	<value_PROJECT_ID> deleted: value is empty string	tc_sa_26.json	{ "PROJECT_ID": "", "FILENAME": "ABcd1234.pdf" }	EnterpriseManagementException: JSON data has no valid values	Node 12 deleted
27	12	NT	DUPLICATION	tc_sa_27_value_pid_duplicated	<value_PROJECT_ID> duplicated: 64-char value	tc_sa_27.json	{ "PROJECT_ID": "0123456789abcdef0123456789abcdef0123456789abcdef0123456789abcdef", "FILENAME": "ABcd1234.pdf" }	EnterpriseManagementException: JSON data has no valid values	Node 12 duplicated
28	14	NT	DELETION	tc_sa_28_label_fn_deleted	<label_FILENAME> deleted: FILENAME key missing	tc_sa_28.json	{ "PROJECT_ID": "0123456789abcdef0123456789abcdef", : "ABcd1234.pdf" }	EnterpriseManagementException: The file is not JSON formatted	Node 14 deleted
29	14	NT	DUPLICATION	tc_sa_29_label_fn_duplicated	<label_FILENAME> duplicated: key repeated	tc_sa_29.json	{ "PROJECT_ID": "0123456789abcdef0123456789abcdef", "FILENAME""FILENAME": "ABcd1234.pdf" }	EnterpriseManagementException: The file is not JSON formatted	Node 14 duplicated
30	14	NT	MODIFICATION	tc_sa_30_label_fn_modified	<label_FILENAME> modified: wrong key name	tc_sa_30.json	{ "PROJECT_ID": "0123456789abcdef0123456789abcdef", "FILE_NAME": "ABcd1234.pdf" }	EnterpriseManagementException: JSON does not have expected structure	Node 14 modified
31	15	NT	DELETION	tc_sa_31_colon_fn_deleted	<equal>(Field2) deleted: colon after FILENAME missing	tc_sa_31.json	{ "PROJECT_ID": "0123456789abcdef0123456789abcdef", "FILENAME" "ABcd1234.pdf"	EnterpriseManagementException: The file is not JSON formatted	Node 15/26 deleted

							}		
32	15	NT	DUPLICATION	tc_sa_32_colon_fn_duplicated	<equal>(Field2) duplicated: :: after FILENAME	tc_sa_32.json	{ "PROJECT_ID": "0123456789abcdef0123456789abcdef" , "FILENAME":: "ABcd1234.pdf" }	EnterpriseManagementException: The file is not JSON formatted	Node 15/26 duplicated
33	15	NT	MODIFICATION	tc_sa_33_colon_fn_modified	<equal>(Field2) modified: = instead of :	tc_sa_33.json	{ "PROJECT_ID": "0123456789abcdef0123456789abcdef" , "FILENAME"= "ABcd1234.pdf" }	EnterpriseManagementException: The file is not JSON formatted	Node 15/26 modified
34	16	NT	DELETION	tc_sa_34_value_fn_deleted	<value_FILENAME> deleted: FILENAME value is empty	tc_sa_34.json	{ "PROJECT_ID": "0123456789abcdef0123456789abcdef" , "FILENAME": "" }	EnterpriseManagementException: JSON data has no valid values	Node 16 deleted
35	16	NT	DUPLICATION	tc_sa_35_value_fn_duplicated	<value_FILENAME> duplicated: filename value repeated	tc_sa_35.json	{ "PROJECT_ID": "0123456789abcdef0123456789abcdef" , "FILENAME": "ABcd1234.pdfABcd1234.pdf" }	EnterpriseManagementException: JSON data has no valid values	Node 16 duplicated
36	18	NT	DELETION	tc_sa_36_string_token_deleted	<STRING_TOKEN> deleted: label quotes missing entirely	tc_sa_36.json	{ PROJECT_ID: "0123456789abcdef0123456789abcdef" , "FILENAME": "ABcd1234.pdf" }	EnterpriseManagementException: The file is not JSON formatted	Node 18/30 deleted
37	18	NT	MODIFICATION	tc_sa_37_string_token_modified	<STRING_TOKEN> modified: label is wrong string	tc_sa_37.json	{ "project_id": "0123456789abcdef0123456789abcdef" , "FILENAME": "ABcd1234.pdf" }	EnterpriseManagementException: JSON does not have expected structure	Node 18/30 modified (lowercase)
38	21	NT	DELETION	tc_sa_38_hex_0_chars	<HEX> deleted: PROJECT_ID value	tc_sa_38.json	{ "PROJECT_ID":	EnterpriseManagementException: JSON	Node 21: 0 iterations

					empty (0 chars)		"", "FILENAME": "ABcd1234.pdf" }	data has no valid values	
39	21	NT	DUPLICATION	tc_sa_39_hex_1_char	<HEX> duplicated: 1 char only in PROJECT_ID	tc_sa_39.json	{ "PROJECT_ID": "a", "FILENAME": "ABcd1234.pdf" }	EnterpriseManagementException: JSON data has no valid values	Node 21: 1 iteration
40	31	T	DELETION	tc_sa_40_hex_31_chars	Hex terminal: 31 chars (n-1 boundary)	tc_sa_40.json	{ "PROJECT_ID": "0123456789abcdef0123456789abcde", "FILENAME": "ABcd1234.pdf" }	EnterpriseManagementException: JSON data has no valid values	Node 31: 31 iters (BVNV)
41	31	T	VALID	tc_sa_41_hex_32_chars	Hex terminal: 32 chars (valid boundary)	tc_sa_41.json	{ "PROJECT_ID": "0123456789abcdef0123456789abcdef", "FILENAME": "ABcd1234.pdf" }	Returns SHA-256 hash: c29b7d03db0e597a9a178318e0121697c8a2523c874190c1d5cfb352e0dcbd55 (use @freeze_time in test)	Node 31: 32 iters (BVV)
42	31	T	DUPLICATION	tc_sa_42_hex_33_chars	Hex terminal: 33 chars (n+1 boundary)	tc_sa_42.json	{ "PROJECT_ID": "0123456789abcdef0123456789abcdefa", "FILENAME": "ABcd1234.pdf" }	EnterpriseManagementException: JSON data has no valid values	Node 31: 33 iters (BVNV)
43	31	T	MODIFICATION	tc_sa_43_hex_nonhex_char	Hex terminal: non-hex char (g) in value	tc_sa_43.json	{ "PROJECT_ID": "0123456789abcdef0123456789abcdeg", "FILENAME": "ABcd1234.pdf" }	EnterpriseManagementException: JSON data has no valid values	Node 31: invalid hex char
44	31	T	MODIFICATION	tc_sa_44_hex_symbol_char	Hex terminal: symbol (@) in PROJECT_ID	tc_sa_44.json	{ "PROJECT_ID": "0123456789abcdef0123456789abcde@", "FILENAME": "ABcd1234.pdf" }	EnterpriseManagementException: JSON data has no valid values	Node 31: symbol in hex
45	24	NT	DELETION	tc_sa_45_fn_token_deleted	<FILENAME_TOKEN> deleted: label quotes	tc_sa_45.json	{ "PROJECT_ID":	EnterpriseManagementException: The	Node 24/32 deleted

					missing		"0123456789abcdef0123456789abcdef", FILENAME: "ABcd1234.pdf" }	file is not JSON formatted	
46	24	NT	MODIFICATION	tc_sa_46_fn_token_modified	<FILENAME_TOKEN> modified: wrong label string	tc_sa_46.json	{ "PROJECT_ID": "0123456789abcdef0123456789abcdef", "filename": "ABcd1234.pdf" }	EnterpriseManagementException: JSON does not have expected structure	Node 24/32 modified (lowercase)
47	28	NT	DELETION	tc_sa_47_fn_value_deleted	<FILENAME_VALUE> deleted: only extension remains	tc_sa_47.json	{ "PROJECT_ID": "0123456789abcdef0123456789abcdef", "FILENAME": ".pdf" }	EnterpriseManagementException: JSON data has no valid values	Node 28: NAME part missing
48	28	NT	DUPLICATION	tc_sa_48_fn_value_duplicated	<FILENAME_VALUE> duplicated: name+ext repeated	tc_sa_48.json	{ "PROJECT_ID": "0123456789abcdef0123456789abcdef", "FILENAME": "ABcd1234.pdfABcd1234.pdf" }	EnterpriseManagementException: JSON data has no valid values	Node 28 duplicated
49	33	NT	DELETION	tc_sa_49_name_deleted	<NAME> deleted: no name, only extension	tc_sa_49.json	{ "PROJECT_ID": "0123456789abcdef0123456789abcdef", "FILENAME": ".pdf" }	EnterpriseManagementException: JSON data has no valid values	Node 33 deleted
50	33	NT	DUPLICATION	tc_sa_50_name_duplicated	<NAME> duplicated: 16-char name	tc_sa_50.json	{ "PROJECT_ID": "0123456789abcdef0123456789abcdef", "FILENAME": "ABcd1234ABcd1234.pdf" }	EnterpriseManagementException: JSON data has no valid values	Node 33 duplicated → 16 chars
51	34	NT	DELETION	tc_sa_51_ext_deleted	<EXTENSION> deleted: no extension in filename	tc_sa_51.json	{ "PROJECT_ID": "0123456789abcdef	EnterpriseManagementException: JSON data has no valid	Node 34 deleted

							0123456789abcdef" , "FILENAME": "ABcd1234" }	values	

52	34	NT	DUPLICATION	tc_sa_52_ext_duplicated	<EXTENSION> duplicated: .pdf.pdf	tc_sa_52.json	{
							"PROJECT_ID":
							"0123456789abcdef
							0123456789abcdef"
							,
							"FILENAME":
							"ABcd1234.pdf.pdf
							"
							}
								EnterpriseManagementException: JSON data has no valid values	Node 34 duplicated

53	34	NT	MODIFICATION	tc_sa_53_ext_invalid	<EXTENSION> modified: .txt (not allowed)	tc_sa_53.json	{
							"PROJECT_ID":
							"0123456789abcdef
							0123456789abcdef"
							,
							"FILENAME":
							"ABcd1234.txt"
							}
								EnterpriseManagementException: JSON data has no valid values	Node 34 modified to .txt

54	35	NT	DELETION	tc_sa_54_alphanum_0_chars	<ALPHANUMERIC> deleted: empty NAME (0 chars)	tc_sa_54.json	{
							"PROJECT_ID":
							"0123456789abcdef
							0123456789abcdef"
							,
							"FILENAME":
							".pdf"
							}
								EnterpriseManagementException: JSON data has no valid values	Node 35: 0 iterations

55	35	NT	DUPLICATION	tc_sa_55_alphanum_1_char	<ALPHANUMERIC> duplicated: 1-char NAME	tc_sa_55.json	{
							"PROJECT_ID":
							"0123456789abcdef
							0123456789abcdef"
							,
							"FILENAME":
							"A.pdf"
							}
								EnterpriseManagementException: JSON data has no valid values	Node 35: 1 iteration

56	37	T	DELETION	tc_sa_56_alphanum_7_chars	Alphanum terminal: 7 chars (n-1 boundary)	tc_sa_56.json	{
							"PROJECT_ID":
							"0123456789abcdef
							0123456789abcdef"
							,
							"FILENAME":
							"ABcd123.pdf"
							}
								EnterpriseManagementException: JSON data has no valid values	Node 37: 7 iters (BVNV)

57	37	T	VALID	tc_sa_57_alphanum_8_chars	Alphanum terminal: 8 chars (valid boundary)	tc_sa_57.json	{ "PROJECT_ID": "0123456789abcdef 0123456789abcdef" ,	Returns SHA-256 hash: 4f39a6d7c82f1dd72850f1abe960f868dfea67dd917781d38f9	Node 37: 8 iters (BVV)

							"FILENAME": "ABcd1234.pdf" }	f4d565f4dc47b (use @freeze_time in test)	
58	37	T	DUPLICATION	tc_sa_58_alphanum_9_chars	Alphanum terminal: 9 chars (n+1 boundary)	tc_sa_58.json	{ "PROJECT_ID": "0123456789abcdef0123456789abcdef", "FILENAME": "ABcd12345.pdf" }	EnterpriseManagementException: JSON data has no valid values	Node 37: 9 iters (BVNV)
59	37	T	MODIFICATION	tc_sa_59_alphanum_symbol	Alphanum terminal: symbol (%) in NAME	tc_sa_59.json	{ "PROJECT_ID": "0123456789abcdef0123456789abcdef", "FILENAME": "ABcd123%.pdf" }	EnterpriseManagementException: JSON data has no valid values	Node 37: symbol char
60	37	T	MODIFICATION	tc_sa_60_alphanum_space	Alphanum terminal: space in NAME	tc_sa_60.json	{ "PROJECT_ID": "0123456789abcdef0123456789abcdef", "FILENAME": "ABcd 234.pdf" }	EnterpriseManagementException: JSON data has no valid values	Node 37: space in NAME
61	38	T	VALID	tc_sa_61_ext_pdf	Extension terminal: .pdf (valid)	tc_sa_61.json	{ "PROJECT_ID": "0123456789abcdef0123456789abcdef", "FILENAME": "ABcd1234.pdf" }	Returns SHA-256 hash: c29b7d03db0e597a9a178318e0121697c8a2523c874190c1d5cfb352e0dcbd55 (use @freeze_time in test)	Node 38: .pdf
62	39	T	VALID	tc_sa_62_ext_docx	Extension terminal: .docx (valid)	tc_sa_62.json	{ "PROJECT_ID": "0123456789abcdef0123456789abcdef", "FILENAME": "ABcd1234.docx" }	Returns SHA-256 hash: a164c3ee96851be75072cc0a2ed8e0b94059e55b8378e33a9115d530db3d2bdf (use @freeze_time in test)	Node 39: .docx
63	40	T	VALID	tc_sa_63_ext_xlsx	Extension terminal: .xlsx (valid)	tc_sa_63.json	{ "PROJECT_ID": "0123456789abcdef0123456789abcdef", "FILENAME": "ABcd1234.xlsx" }	Returns SHA-256 hash: 352bf7fec722f6077a56e54d7e9118a463e6814b6c74b2235e665a299319ec21 (use @freeze_time in test)	Node 40: .xlsx

64	38	T	MODIFICATION	tc_sa_64_ext_uppercase	Extension terminal: .PDF (uppercase, not allowed)	tc_sa_64.json	{ "PROJECT_ID": "0123456789abcdef0123456789abcdef", "FILENAME": "ABcd1234.PDF" }	EnterpriseManagementException: JSON data has no valid values	Node 38 modified: case
65	36	NT	MODIFICATION	tc_sa_65_ext_csv	<EXTENSION_TOKEN> modified: .csv (not allowed)	tc_sa_65.json	{ "PROJECT_ID": "0123456789abcdef0123456789abcdef", "FILENAME": "ABcd1234.csv" }	EnterpriseManagementException: JSON data has no valid values	Node 36 modified to .csv

Structural Test Cases

#	PATH	ID TEST	DESCRIPTION	FIELD	EXPECTED RESULT	OBSERVATIONS
1	1→2→15→16	TC-ST-01	Exception raised when opening JSON file	JSON file path	EnterpriseManagementException: Input file not found.	Exception at open JSON file: try block exits before any data is read
2	1→2→3→15→16	TC-ST-02	Exception raised when loading data from JSON file	JSON file content	EnterpriseManagementException:The file is not JSON formatted.	Exception at load data: file opened but data could not be loaded
3	1→2→3→4→15→16	TC-ST-03	Exception raised when extracting project & filename from JSON data	JSON data structure (project/filename fields)	EnterpriseManagementException: JSON does not have the expected structure.	Exception at get project & filename: data loaded but JSON structure invalid
4	1→2→3→4→5(T)→15→16	TC-ST-04	MD5 validation fails (invalid MD5 hash)	MD5 hash value	EnterpriseManagementException: JSON data has no valid values.	Branches at MD5 validation: invalid MD5 takes exception edge
5	1→2→3→4→5(F)→6(T)→15→16	TC-ST-05	MD5 is valid but file extension/type validation fails	File extension / type	EnterpriseManagementException: JSON data has no valid values.	Branches at file extension/type validation: invalid type takes exception edge
6	1→2→3→4→5(F)→6(F)→7→8→9→10→11→12→13→15→16	TC-ST-06	Valid MD5 and file type; exception raised when opening/writing to storage file (node 13)	Storage file write operation	EnterpriseManagementException: Internal processing error when getting the file_signature.	Exception at open & write to storage file: all prior steps succeed but final write fails
7	1→2→3→4→5(F)→6(F)→7→8→9→10(except)→11→12→13→14→16	TC-ST-07	Valid MD5 and file type; FileNotFound exception at node 10 (open storage file), creates empty al_docs, then successfully writes and returns signature	Storage file open (FileNotFoundError path)	Returns file_signature successfully.	FileNotFound exception at open storage file: takes alternate branch, creates empty al_docs variable
8	1→2→3→4→5(F)→6(F)→7→8→9→10(try)→12→13→14→16	TC-ST-08	Valid MD5 and file type; storage file opens successfully; docs appended; signature returned	Storage file (exists and readable)	Returns file_signature successfully.	Nominal path: all try blocks succeed, both validations pass, storage file read and written
9	1→2→3→4→5(F)→6(F)→7→8→9→10(try)→12→13(except)→15→16	TC-ST-09	Valid MD5 and file type; storage file opens but exception raised during open & write (node 13)	Storage file write (after successful open)	EnterpriseManagementException: Internal processing error when getting the file_signature.	Exception at open & write to storage file: storage file opened but write operation fails